The documentation and process conversion measures necessary to comply with this amendment shall be completed by 6 September 1994

INCH-POUND

MIL-S-19500/260D
AMENDMENT 4
6 June 1994
SUPERSEDING
AMENDMENT 3
20 October 1992

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER,
TYPES 1N1202A, 1N1204A, 1N1206A, 1N3671A, 1N3673A, AND AR
JAN, JANTX, JANTXV, AND JANS

This amendment forms a part of MIL-S-19500/260D, dated 21 April 1989 and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* Title, delete and substitute as printed below:

"SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER,
TYPES 1N1202A, 1N1204A, 1N1206A, 1N3671A, 1N3673A,
AND AR VERSIONS, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC"

* 1.1, end of paragraph, add: "Two levels of product assurance for die.".

* 1.3, after "DO-4)" Add: ", see figure 1A for JANHC and JANKC die dimensions."

1.4, I_{FSM} column: Delete "I_{FSM} " and substitute "$i_{f(surge)}$ " .

* Beneficial comments note, delete address and substitute the following:"Commander, Defense Electronics Supply Center, ATTN: DESC-ELDT, 1507 Wilmington Pike, Dayton, OH 45444-5765,".

PAGE 2

* 3.3, line 2: Delete "figure 1" and substitute "figures 1 and 1A".

* Add the following new paragraph:

"4.2.1 JANHC and JANKC devices. Qualification for JANHC and JANKC devices shall be in accordance with appendix H of MIL-S-19500."

Distribution statement A Approved for public release; distribution is unlimited.

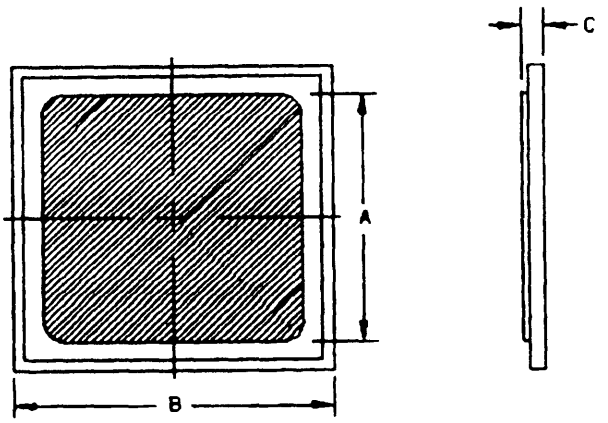
Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:57Z
Check the source to verify that this is the current version before use.

PAGE 3

* Add the following new figure 1A:
"

A Version

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.116	.122	2.95	3.10
B	.147	.153	3.73	3.89
C	.009	.012	.229	.305



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die are: Metallization {front (anode), consists of: Ti thickness = 1,200 Å minimum, Ni thickness = 2,000 Å minimum, Ag thickness = 4,000 Å minimum. Metallization {back, (cathode)} consists of: Ti thickness = 1,200 Å minimum, NI thickness = 2,000 Å minimum, Ag thickness = 4,000 Å minimum.

FIGURE 1A. JANC die dimensions."

PAGE 4

4.3.2: Delete and substitute as follows:

"4.3.1 Thermal response. The thermal response measurements shall be performed in accordance with method 3101 of MIL-STD-750. The thermal response conditions and maximum thermal response limit shall be derived by each vendor. The chosen thermal response measurement and conditions for each device in the qualification lot shall be submitted in the qualification report and a thermal response curve shall be plotted. The chosen thermal response values shall be considered final after the manufacturer has had the opportunity to test five consecutive lots. Heating current (I_H) ≥ rated I_O; t_H = 150 to 400 ms; t_{MD} = 50 to 300 µs; 50 mA ≤ I_M ≤ 250 mA."

PAGE 5

* "4.3.4 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-S-19500, appendix H. As a minimum, die shall be 100-percent probed to ensure compliance with I_{R1} of group A, subgroup 2."

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:57Z
Check the source to verify that this is the current version before use.

PAGE 6

TABLE I, subgroup 2: Delete the following:

Forward voltage	4011	I_F = if(surge) t_p = 800 µs		V_{F1}		2.5	V dc

TABLE I, add the following.

Subgroup 6							
Forward voltage 2/	4011	I_F = if(surge) t_p = 800 µs		V_{F1}		2.5	V dc
Forward voltage 2/	4011	I_F = If(surge) t_p = 8.3 ms		V_{F1}		2.3	V dc

2/ V_{F1} shall be performed with either t_p = 800 µs or t_p 8.3 ms".

PAGE 8

TABLE IIa, subgroup 6, thermal resistance, conditions column: Delete and substitute "$R_{\theta JC}$ = 2.0°C/W maximum; I_M = 50 mA to 250 mA; $I_H \geq$ rated I_O; t_H = 150 to 400 ms; t_{MD}= 50 to 300 µs".

PAGE 12

TABLE IV, subgroup 3, DPA, method column: Add "2101".
TABLE IV, subgroup 4, thermal resistance, conditions column: Delete and substitute "$R_{\theta JC}$ = 2.0°C/W maximum; $I_M \geq$ 100 mA; $I_H \geq$ rated I_O; t_H = 150 to 400 ms; t_{MD}= 50 to 300 µs".

PAGE 13

TABLE V, delete table title and substitute: "TABLE V. Groups A, B, C, and E electrical end-point measurements. "; Also, in step 4, thermal response, conditions column,add the following: "$R_{\theta JC}$ = 2.0°C/W maximum; $I_M \geq$ 100 mA; $I_H \geq$ rated I_O; t_H = 150 to 400 ms; t_{MD}= 50 to 300 µs".

PAGE 14

6.2, add the following: "d. For die acquisition, the JANHC or JANKC letter version shall be specified (see figure 1A)."

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:57Z
Check the source to verify that this is the current version before use.

* 6.3, delete in it's entirety, including table, and substitute the following new paragraph and table:

" 6.3 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example, JANHCA1N2104A) will be identified on the QPL.

JANHC and JANKC ordering information	
	Manufacturer
PIN	59377
	1/
1N1202A	JANHCA1N1202A
1N1204A	JANHCA1N1204A
1N1206A	JANHCA1N1206A
1N3671A	JANHCA1N3671A
1N3673A	JANHCA1N3673A

1/ For JANKC, replace the "JANHC" suffix with "JANKC"."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - AS, CG, MC
Air Force - 19, 85, 99

Preparing activity:
DLA - ES

(Project 5961-1504)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:57Z
Check the source to verify that this is the current version before use.